ArtiAtlas: A Holistic Ecosystem for Diverse 3D Articulation Modeling

JUNZHE LU, Tsinghua University, China and ByteDance Seed, China
JING LIN*, Nanyang Technological University, Singapore and ByteDance Seed, China
ZIANG CAO, Nanyang Technological University, Singapore
HAOQIAN WANG†, Tsinghua University, China
ZIWEI LIU, Nanyang Technological University, Singapore
JIANFENG ZHANG†, ByteDance Seed, China

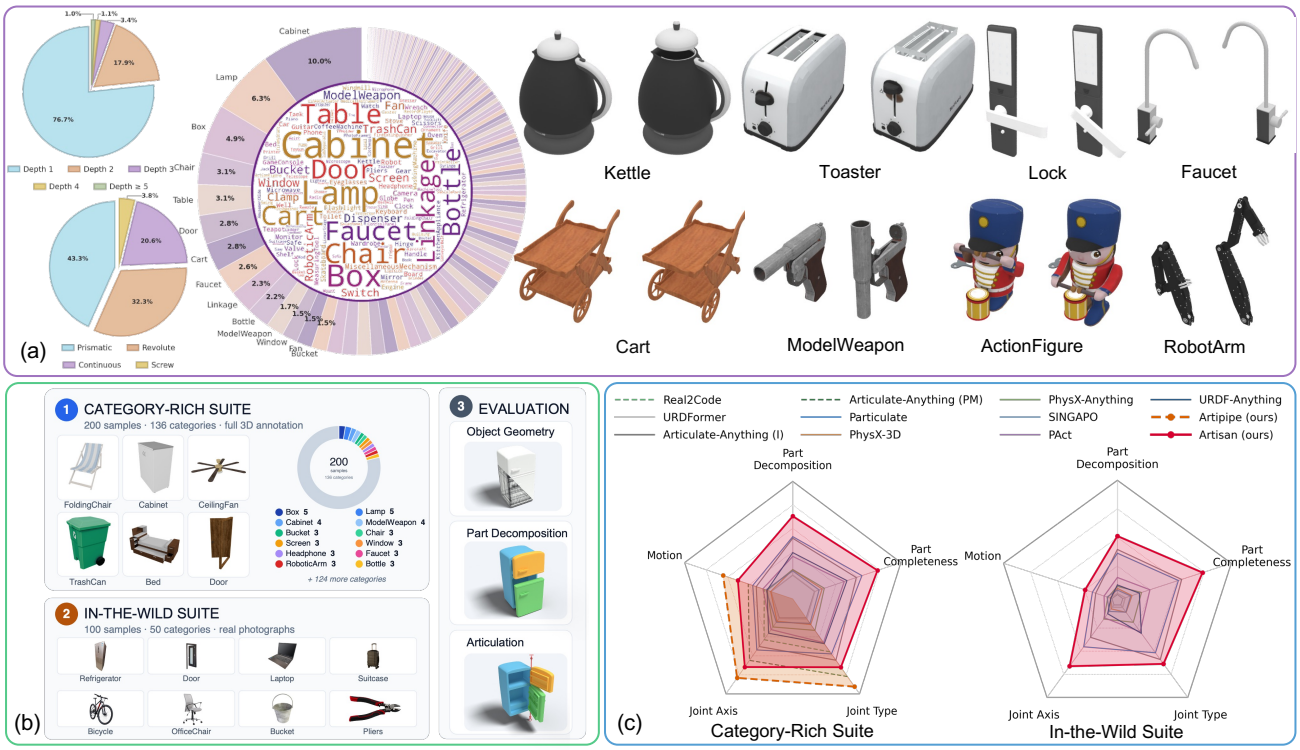


Fig. 1. **A holistic ecosystem for diverse 3D articulation modeling.** (a) ArtiAtlas, our dataset of 14,556 articulated 3D assets across 173 categories with full kinematic annotation: the inner pie charts summarise the kinematic-tree depth and joint-type distributions, the donut surfaces the category long tail. (b) ArtiBench, a unified evaluation suite that pairs a Category-Rich Suite of 200 ArtiAtlas assets with full 3D references and an In-the-Wild Suite of 100 real photographs across 50 categories without 3D references, and grades every method along the same three axes: object geometry, part decomposition, and articulation. (c) Artisan, our query-based whole-mesh-to-URDF model trained on ArtiAtlas, dominates prior baselines across all five VLM-judged dimensions on both suites; the part-mesh-input ArtiPipe that produces the dataset annotations is plotted alongside to indicate the headroom that remains.

*Project lead.
†Corresponding authors.

Authors' Contact Information: Junzhe Lu, Tsinghua University, Shenzhen, China and ByteDance Seed, Beijing, China, junzhelu.bh@gmail.com; Jing Lin, Nanyang Technological University, Singapore, Singapore and ByteDance Seed, Beijing, China, jinglin.stu@gmail.com; Ziang Cao, Nanyang Technological University, Singapore, Singapore, ziang001@e.ntu.edu.sg; Haoqian Wang, Tsinghua University, Shenzhen, China, wanghaoqian@tsinghua.edu.cn; Ziwei Liu, Nanyang Technological University, Singapore, Singapore, zwliu.hust@gmail.com; Jianfeng Zhang, ByteDance Seed, Beijing, China, jf.zhang958@outlook.com.

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The creation of interactable, articulated 3D assets is a cornerstone of modern computer graphics and physical simulation. Progress in articulation modeling is currently bottlenecked by limited data scale, inconsistent evaluation protocols, and representational fragility under long-tail kinematic chains. We present three complementary contributions to address these gaps: *ArtiAtlas* is a large-scale dataset of 14,556 annotated articulated objects across 173 categories, constructed via a scalable human-machine pipeline for semantic-part consolidation and kinematic annotation. *ArtiBench* is a unified evaluation suite pairing an in-distribution split of ArtiAtlas with an image-conditioned in-the-wild set, and introducing calibrated VLM-based metrics that align

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closely with human preference where ground-truth 3D annotation is unavailable. *Artisan* is a part-aware articulation model with a set-prediction query interface, mask-aware bidirectional decoding, and geometry-anchored axis supervision, trained end-to-end to predict variable-cardinality link sets directly from raw meshes. Under a strict per-joint success criterion, all evaluated baselines achieve below 10% on ArtiAtlas; Artisan raises this to 30% and clearly leads on the in-the-wild VLM scores, yet no method—ours included—is close to reliable fine-grained kinematic prediction. We release the 10,005-object public subset of ArtiAtlas, ArtiBench, and Artisan as a shared foundation for future work on realistic articulation understanding and generation.

CCS Concepts: • **Computing methodologies** → **Computer vision**; *Shape modeling*.

Additional Key Words and Phrases: articulated 3D objects, dataset, benchmark, motion annotation

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1 Introduction

Interactable, articulated 3D assets underpin a broad range of applications, from physically grounded simulation [Xiang et al. 2020] and robotic manipulation [Geng et al. 2023; Liu et al. 2022] to interactive digital environments [Wang et al. 2024]. Despite substantial advances in static geometry generation [Hunyuan3D et al. 2025; Xiang et al. 2025a,b], modeling the kinematic structure of real-world objects remains a largely unsolved problem.

Progress in 3D articulation modeling is currently bottlenecked along three dimensions. First, training data is largely confined to narrow categories and simplified motions [Geng et al. 2023; Liu et al. 2022; Xiang et al. 2020], preventing models from acquiring sufficiently broad priors. Second, existing benchmarks [Iliash et al. 2026a; Li et al. 2025; Liu et al. 2025a] are concentrated in a small set of categories, which can obscure failures on fine-grained articulation. Third, current methods [Li et al. 2025, 2026; Liu et al. 2026] exhibit representational fragility: they overfit to constrained laboratory settings and degrade sharply on objects with long-tail part counts, small components, or non-standard kinematic chains.

To address these limitations concurrently, we introduce three contributions. *ArtiAtlas* is the largest 3D articulation dataset to date, comprising 14,556 objects across 173 categories, constructed via a scalable human-machine pipeline, *ArtiPipe*, that automates VLM-based kinematic inference and geometric axis estimation, while retaining human review for quality control, reducing per-asset annotation cost without sacrificing label precision (Figure 1(a)). Large-scale data alone, however, cannot drive progress without a consistent evaluation protocol. *ArtiBench* fills this gap by harmonizing metrics across three input modalities and organizing assessment into object geometry, part decomposition, and articulation quality; it further introduces an image-conditioned in-the-wild set evaluated by a calibrated VLM judge validated to reach sample-level Spearman $\rho \geq 0.80$ against human scores (Figure 1(b)). To establish a strong baseline under this protocol, we develop *Artisan*, a part-aware mesh-to-URDF model with a set-prediction query interface that gracefully handles variable-cardinality parts, a mask-aware bidirectional decoder for precise part boundary reasoning, and a geometry-anchored axis-pivot loss that directly supervises revolute axis placement on each predicted link (Figure 1(c)).

Table 1. Comparison of related articulated-object and 3D-part datasets. † Original AKB-48 download is unavailable; per-sample stats use the subset contained in GAPartNet release. ‡ Articulated subset reported (rigid items in the same release are excluded). § VLM-generated free-form labels. ★ Counts from paper; per-sample stats use the released subset (3,543 assets).

Dataset	Decomp.	#Samples	#Cats	Parts (μ/max)	Depth (μ/max)	Year
PartNet [Mo et al. 2019]	connectivity	26,671	24	21.5 / 158	–	2019
PartNeXt [Wang et al. 2025]	semantics	23,519	50	14.9 / 4,924	–	2025
PartVerse [Dong et al. 2025]	semantics	12,030	175	7.6 / 139	–	2025
PartVerse-XL [Ding et al. 2025]	semantics	40,000	>200	8.0 / 198	–	2025
PartNet-Mobility [Xiang et al. 2020]	kinematics	2,347	46	6.00 / 116	1.14 / 6	2020
AKB-48† [Liu et al. 2022]	kinematics	2,037	48	2.12 / 4	1.0 / 1	2022
GAPartNet [Geng et al. 2023]	kinematics	1,166	27	7.62 / 116	1.05 / 2	2023
GRScenes-100‡ [Wang et al. 2024]	kinematics	1,770	25	3.06 / 14	1.00 / 2	2024
PhysXNet‡ [Cao et al. 2025a]	kinematics	9,328	274§	5.16 / 126	1.07 / 4	2025
Artiverse★ [Iliash et al. 2026b]	kinematics	5,402	88	5.58 / 118	1.14 / 5	2026
ArtiAtlas (ours)	kinematics	14,556	173	5.76 / 209	1.33 / 12	2026

Extensive evaluation yields a sobering picture. Under a strict end-to-end success criterion, all evaluated baselines, run in their released configurations, achieve below 10% success on ArtiAtlas. Artisan raises this to 30% and attains clearly stronger VLM scores on in-the-wild inputs, yet all methods, including ours, remain far from reliable prediction of fine-grained kinematic parameters. We view the clarity of this diagnosis as itself a contribution, and offer ArtiAtlas, ArtiBench, and Artisan as a shared foundation for future work on realistic articulation understanding and generation.

In summary, our core contributions include:

- **ArtiAtlas:** A large-scale dataset of 14,556 articulated objects across 173 categories, built via a scalable human-machine pipeline that extends well beyond household categories.
- **ArtiBench:** A unified evaluation suite spanning multiple input modalities, with calibrated VLM-based metrics that enable reference-free assessment of functional plausibility and expose failure modes obscured by prior benchmarks.
- **Artisan:** A part-aware mesh-to-URDF model that establishes a new state-of-the-art on ArtiBench, while our benchmarking results collectively define how much headroom remains in fine-grained articulation.

2 Related Work

2.1 Articulated Object Modeling

Methods for producing articulated 3D assets differ primarily in input modality. Category-, text-, or structure-conditioned generative models [Chen et al. 2025a; Gao et al. 2025; Lei et al. 2023; Liu et al. 2024; Su et al. 2025] decode part layouts and joint parameters but are largely constrained to household vocabularies. Instance-focused reconstruction methods [Chen et al. 2025b; Jiang et al. 2022a; Liu et al. 2023, 2025b; Shen et al. 2025; Song et al. 2024] recover higher-fidelity assets but strictly require observing target instances in motion. Image-conditioned systems [Cao et al. 2025a,b, 2026; Chen et al. 2024; Jin et al. 2025; Le et al. 2025; Liu et al. 2025a, 2026; Lu et al. 2025] pursue broader accessibility from a single view, though fully

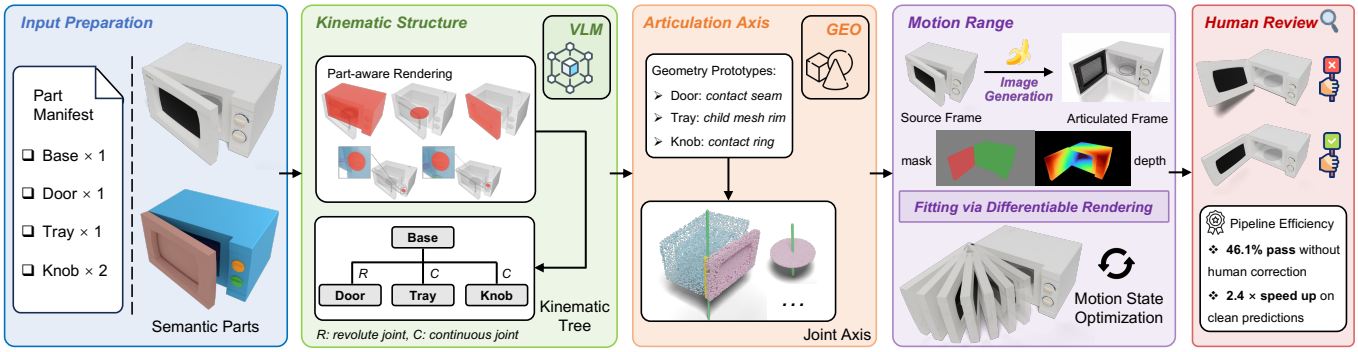


Fig. 2. **Overview of ArtiPipe.** Starting from a part manifest and the corresponding semantic-part-colored mesh, the pipeline factorises articulation annotation into three predictions. *Kinematic Structure* renders part-aware views and queries a VLM for each part's parent and joint type. *Articulation Axis* executes a predefined catalogue of geometric prototypes and selects among them with a second VLM, returning a 3D axis per joint. *Motion Range* edits the rest-state render into an articulated target frame with an image generator, then fits the scalar joint state by differentiable rendering against silhouette and depth losses (VLM fallback is omitted for clarity). Every predicted URDF then enters a two-pass *Human Review* before release; 46.1% of reviewed joints require no axis correction, giving a ∼2.4× review-time speedup on the clean predictions.

hidden parts remain ambiguous and fixed mesh libraries limit long-tail coverage.

Inferring articulation from 3D geometry is a long-standing problem [Geng et al. 2023; Goyal et al. 2025; Jiang et al. 2022b; Wang et al. 2019; Yan et al. 2019; Yu et al. 2024]; a more recent line of work decouples geometry acquisition from articulation by operating directly on static 3D meshes or point clouds [Li et al. 2025, 2026], the setting addressed by our proposed *Artisan* model (Section 4). As strong image-to-3D generators [Hunyuan3D et al. 2025; Xiang et al. 2025b] provide increasingly clean meshes, this paradigm allows articulation models to bypass 2D occlusion ambiguities. However, existing realizations remain capacity-limited: URDF-Anything [Li et al. 2026] and Particulate [Li et al. 2025] rely on restricted point-sampling budgets and rigid output bounds. These constraints degrade resolution on fine geometry and rich kinematic chains—precisely the "hard cases" *ArtiBench* (Section 5) is designed to surface.

2.2 Articulated Object Datasets and Benchmarks

Most recent models train and evaluate on PartNet-Mobility [Xiang et al. 2020] and related scene-level resources [Geng et al. 2023; Liu et al. 2022; Wang et al. 2024]. While foundational, these resources concentrate on limited household categories, preventing models from acquiring broad priors for open-world diversity. Conversely, semantic-only part datasets [Dong et al. 2025; Mo et al. 2019; Wang et al. 2025] are massive but lack joint annotations, which remain expensive to obtain: Motion Annotation Programs transfers expert-authored interactive rules across a shape collection [Xu et al. 2020], and PhysXNet [Cao et al. 2025a] fits candidate planes for an annotator to select, scaling physics-annotated assets but introducing noticeable joint-axis noise in long-tail categories. Artiverse [Iliash et al. 2026b] adds completed interiors and physical attributes but spans only 5,402 objects across 88 categories. We overcome these limitations with ArtiAtlas, the largest and most diverse 3D articulation dataset to date, annotated by ArtiPipe, which predicts the kinematic tree, joint type, axis, and range automatically and leaves humans only to review the annotations (Section 3).

Existing benchmarks inherit similar category biases. A commonly used 77-object PartNet-Mobility evaluation split, used by SINGAPO [Liu et al. 2025a] and later comparisons [Jin et al. 2025; Li et al. 2025; Liu et al. 2026; Wu et al. 2025], covers only seven categories and is severely imbalanced (55 StorageFurniture objects, 10 Tables, and at most four objects in any other category), so aggregate scores are dominated by cuboidal cabinet doors. Recent work attempts to expose generalisation failures with curated out-of-distribution evaluations, such as a 243-asset Lightwheel set across 14 categories [Li et al. 2025] and the 354-object Articulated Containers Dataset introduced with S2O [Iliash et al. 2026a], from which SINGAPO defined a 135-object subset [Liu et al. 2025a] later used by ArtiLatent [Chen et al. 2025a]; both remain concentrated on furniture and kitchen objects, and each line of work uses its own metric protocol, complicating cross-paper comparison. Our proposed ecosystem resolves these gaps: ArtiAtlas provides the necessary open-world distribution, while *ArtiBench* establishes a unified evaluation protocol (Section 5) applied identically across all models to ensure unbiased comparison.

3 Dataset

3.1 Dataset Overview and Representation

We introduce **ArtiAtlas**, a large-scale dataset of articulated 3D assets with broad category coverage, diverse kinematics, and annotations that pair a part segmentation with joint parameters. With 14,556 assets spanning 173 categories, ArtiAtlas expands the scope of kinematics data.

Each asset is represented as $O = (\mathcal{L}, \mathcal{T}, \Phi)$. The link decomposition groups raw geometry into rigid functional links $\mathcal{L} = \{l_0, \ldots, l_n\}$, each with a semantic label. The kinematic annotation specifies their relative motion through a rooted kinematic tree $\mathcal{T}$ and articulation parameters Φ. We support fixed, revolute, prismatic, continuous, and screw joints. For each joint, Φ includes the joint type τ, motion axis $\mathbf{u}$, origin or pivot $\mathbf{q}$, and feasible range $[r_{min}, r_{max}]$. The link decomposition defines the geometric target, while the kinematic annotation defines the structural validity of motion.

3.2 Data Sourcing and Semantic Consolidation

We construct ArtiAtlas from three source groups: (i) the multi-source raw-asset stream (9,721 objects from ten public research datasets and 1,287 from a commercial partner); (ii) PartNeXt (1,827); and (iii) PartVerse-XL (1,721). The raw-asset stream undergoes visual-semantic screening, geometry checks, and balanced sampling, while the public sources enter after these early stages because they already provide part decompositions and semantic labels. Filtering funnels and retention statistics are reported in Supplementary Sec. B. All VLM-assisted stages use Gemini-3-Flash with a single multimodal few-shot example; prompts are listed in Supplementary Sec. I.

Neither source type is segmented at the granularity articulation requires: the raw-asset stream is split by mesh connectivity, far finer than its rigid links, while the semantic parts of PartNeXt and PartVerse-XL follow labels rather than motion, separating a cabinet door from its handle although the two move as one rigid body. Annotators therefore use a custom Blender interface to merge low-level fragments into functional rigid parts and discard assets that are under-segmented or structurally unusable. The resulting per-object semantic part manifest is the input to our articulation annotation pipeline. Interface details, examples, and curation criteria are given in Supplementary Sec. C.

3.3 Articulation Annotation Pipeline

ArtiPipe, our articulation annotation pipeline, consumes the prepared part manifests and decomposes URDF construction into three predictions (Figure 2). Kinematic structure is guided by commonsense regularities, axis estimation by geometric evidence, and motion range by physical scale plus a consistent sign convention.

First, a VLM predicts each part's parent and joint type from rendered contexts and the semantic manifest, relying on physical priors such as drawers sliding and cabinet doors hinging rather than fine-grained geometric fitting. On PartNet-Mobility, this stage reaches 96.9% parent-assignment accuracy and 96.0% joint-type accuracy; details are provided in Supplementary Sec. D.1.

Second, axes are estimated with a geometry-first, VLM-selected procedure. A catalogue of prototypes proposes candidate axes from part-intrinsic geometry or child–parent contact regions, and a second VLM selects the most plausible one under physical priors. Under strict PartNet-Mobility criteria, this stage reaches 80.5% axis accuracy; an oracle selector raises this to 86.1%, indicating broad prototype coverage with residual errors from selector confusion and out-of-catalogue cases. Supplementary Sec. D.2 provides the catalogue and failure analysis.

Finally, motion ranges are estimated with a selective visual-generation path and VLM fallback. The primary path edits a rest-state render into a near-limit target frame, then recovers the scalar joint state by differentiable render-and-fit; image editing is used instead of video generation to preserve camera parameters. If a joint is not selected for the visual path, or an automatic critic rejects the synthesized frame, the VLM fallback resolves the sign ambiguity using an explicit axis-direction cue. All predicted URDFs then undergo a two-pass Blender review. Appendices detail the fitting objective, review interface, and per-stage accuracy. In final review, 46.1% of joints require no axis correction, substantially reducing effort relative to fully manual kinematic fitting.

3.4 Coverage and Kinematic Complexity

As summarized in Table 1, ArtiAtlas expands articulated-object data in both scale and diversity. It covers domestic objects, industrial tools and mechanisms, and consumer items, with long-tail categories such as locks, robot arms, action figures, keyboards, and model weapons (Figure 1(a)); taxonomy definitions and complete per-category counts are given in Supplementary Sec. E.

The kinematic structures are also diverse. Assets contain 5.76 parts on average and up to 209 in the largest case; the average chain depth is 1.33, with a tail reaching depth 12. Revolute and prismatic joints dominate, while continuous and screw joints capture less common motion types. Although most assets have shallow trees typical of man-made objects, the deeper tail is important because current methods often degrade once motion dependencies extend beyond a single independent joint. This makes ArtiAtlas both a demanding training set and a diagnostic evaluation platform.

4 Articulation Model

ArtiAtlas turns whole-mesh articulation into a supervised prediction problem where the number of links varies per object and is not bounded by a small, category-dependent constant. Artisan is designed for this regime, taking a single static mesh as input and exporting a complete URDF.

The central modeling challenge is variable cardinality. Particulate [Li et al. 2025] represents parts with a fixed bank of part queries but normalizes point-to-part assignments over all query slots, so unmatched slots still participate in the softmax as the slot budget grows. We instead adopt a DETR-style set-prediction contract with an explicit no-object state [Carion et al. 2020], combined with the independent per-query masks of mask-classification transformers [Cheng et al. 2022; Schult et al. 2023]. This gives unused slots a supervised opt-out path while preserving a fixed-size neural output.

4.1 Problem Definition

Given a mesh $\mathcal{M} = (V, F)$, Artisan predicts a segmentation $S : F \to [P]$ into an unknown number P of rigid links, a rooted kinematic tree K, and joint labels with continuous motion parameters on every tree edge. Prismatic joints are parameterized by a translation direction and scalar range, revolute joints by an axis line and angular range, and screw joints reuse both parameterizations. Following Particulate [Li et al. 2025], continuous joints are encoded during training as revolute joints with a 360° range.

4.2 Point and Query Representation

Artisan operates on sampled surface points rather than mesh faces. We sample N points with coordinates, normals, and frozen PartField features [Liu et al. 2025c], and project them into point tokens $X \in \mathbb{R}^{N\times D}$. A learned set of M part queries $Q \in \mathbb{R}^{M\times D}$ represents candidate links. For query j, the per-point mask logit $\hat{s}_{ji}$ is computed by a dot product between projected query and point features.

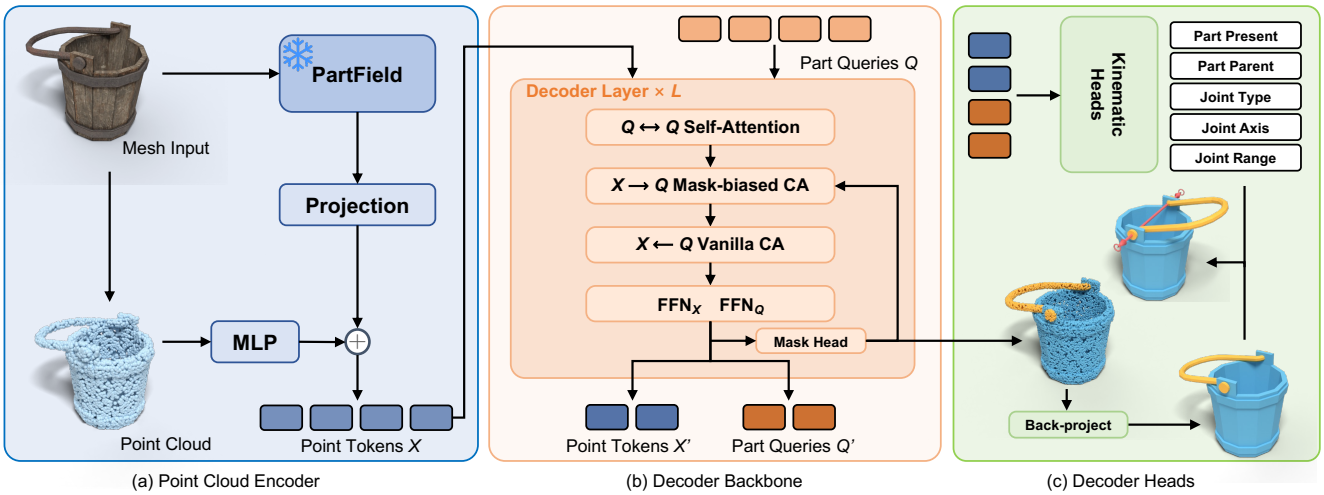


Fig. 3. **Artisan overview.** (a) The input mesh is encoded by a frozen PartField backbone into per-point features that are projected and combined with positional/normal embeddings to form point tokens X. (b) Part queries Q and point tokens are refined by L bidirectional decoder layers; each layer applies query self-attention, mask-biased $X \to Q$ cross-attention, vanilla $X \leftarrow Q$ cross-attention, and parallel FFNs. (c) The refined queries and point tokens feed parallel kinematic heads; per-point mask logits are back-projected to mesh faces to recover the link decomposition, which is assembled into a URDF.

4.3 Mask-Aware Bidirectional Decoder

The decoder jointly refines point tokens and part queries through a stack of attention layers. Each layer updates queries with self-attention and query-to-point cross-attention, and updates point tokens with point-to-query cross-attention. This bidirectional update lets candidate parts read dense geometry while letting point tokens be conditioned on the current link hypotheses.

We make query-to-point attention mask-aware: after each layer predicts intermediate masks, the next layer restricts query j to points with $\sigma(\hat{s}_{ji}^{\ell-1}) \geq 0.5$ by assigning zero attention bias inside the mask and $-\infty$ outside. The first layer and empty-mask queries fall back to global attention. The reverse $X \leftarrow Q$ direction remains unmasked, so boundary and currently misassigned points can still receive information from all candidate links.

4.4 Output Heads

From each final query, parallel heads predict present/no-object classification, per-point mask logits, parent scores over ordered query pairs, and joint type. We follow the URDF-compatible motion representation of Particulate [Li et al. 2025]: revolute joints predict an axis direction and angular range, prismatic joints predict a translation direction and range, and screw joints use both heads.

For revolute joints, each point assigned to a link predicts a 3D point on the revolute axis, and the median of these votes is the exported pivot. Because this per-point target is locally anchored, we add a query-level axis-pivot head supervised by the same median, providing a direct link-level gradient for pivot localization.

4.5 Training

Training follows a DETR-style set-prediction objective. For each object, ground-truth links are Hungarian-matched to distinct predicted queries using present/no-object classification and mask quality:

$$C_{jp} = \lambda_{\text{cls}} C_{jp}^{\text{present}} + \lambda_{\text{bce}} C_{jp}^{\text{bce}} + \lambda_{\text{dice}} C_{jp}^{\text{dice}}. \tag{1}$$

Matched queries receive supervision for masks, tree edges, joint types, axes, ranges, closest-axis votes, and axis-pivot targets, while unmatched queries are trained as no-object. Intermediate decoder layers receive auxiliary class and mask supervision.

The full objective combines set-prediction, tree, joint classification, and continuous motion losses:

$$\mathcal{L} = \sum_t \lambda_t \mathcal{L}_t. \tag{2}$$

Continuous terms use L1 losses on axis directions, ranges, closest-axis votes, and query-level pivots. To remove the sign ambiguity of axis-based joints, we canonicalize each training label by treating

$$(\mathbf{u}, [r_{\min}, r_{\max}]) \equiv (-\mathbf{u}, [-r_{\max}, -r_{\min}]) \tag{3}$$

as equivalent and selecting a deterministic axis sign before deriving motion targets. This avoids bimodal regression targets on axis-based joints. Loss weights and target construction are detailed in Supplementary Sec. F.1.

4.6 Inference and URDF Assembly

At inference, queries above the present-score threshold are retained, with a top-one fallback when none pass. Sampled points are assigned to the retained query with the highest mask logit and projected back to mesh faces. Parent scores between queries are converted into a rooted tree by maximum spanning arborescence, and the predicted joint types, axes, pivots, and ranges complete the URDF. Optional connectivity-based cleanup is described in Supplementary Sec. F.3.

5 Benchmark

ArtiBench is an evaluation protocol that scores generated articulated assets as structured objects with link geometry, a kinematic tree, joint types, joint axes, joint origins, and motion limits, applying the same output contract and metric set to every method. Two suites and three evaluation dimensions cover both controlled and in-the-wild scenarios.

5.1 Benchmark Suites

Category-Rich Suite. 200 held-out ArtiAtlas objects across 136 categories, drawn at random within per-category quotas set by the number of available assets and the prevalence of each category, with full link meshes and kinematic annotations, supporting reference-based geometry and articulation evaluation; the same assets are also rendered into conditioning images for image-input methods. The suite contains only 1.47 objects per category on average, with many categories represented by one or two instances; it therefore emphasizes category breadth rather than intra-category variation and does not support robust per-category estimates.

In-the-Wild Suite. 100 image-conditioned examples spanning 50 categories, two per category. Each sample pairs a real photograph with a foreground mesh from a commercial image-to-3D model (Seed3D 2.0 [ByteDance Seed Team 2026]); the suite has no authoritative 3D reference, so it tests whether a method produces a mechanically plausible and image-faithful asset under real image-to-asset ambiguity.

External Suites. We additionally report the PartNet-Mobility reference evaluation, retained for continuity with prior work, and an evaluation on the 243-object Lightwheel set introduced by Particulate [Li et al. 2025]; both are in Supplementary Secs. G.8 and G.9.

5.2 Tasks and Evaluation Dimensions

Methods fall into three input settings by input–output contract: *part-mesh-to-URDF*, *whole-mesh-to-URDF*, and *image-to-URDF*. All settings share the same output contract, and we report three complementary dimensions. Metric headers use the prefix *R-* for rest-state and *A-* for articulated-state quantities, and *obj*/*part* for whole-object versus per-part scope.

Object Geometry. This dimension measures assembled-shape fidelity. With a 3D reference, all links are assembled at the rest state and compared with whole-object Chamfer distance (R-obj CD). For image-conditioned methods we follow TRELLIS 2.0 [Xiang et al. 2025a] and report image-to-3D alignment via CLIP [Radford et al. 2021], CLIP-N, ULIP-2 [Xue et al. 2024], and Uni3D [Zhou et al. 2024].

Part Decomposition. This dimension measures whether the object is split into functional rigid links. Predicted and reference links are matched by Hungarian assignment with the worst-score penalty for unmatched links [Li et al. 2025], and we report rest-state per-part Chamfer distance and gIoU [Rezatofighi et al. 2019]. Without a 3D reference, we use rest-state self-overlap among predicted link meshes (R-overlap) and VLM scores for decomposition quality and per-part completeness.

Articulation. This dimension measures the predicted kinematic model. Following the type–axis–origin success structure implemented by Articulate-Anything [Le et al. 2025] and related protocols [Iliash et al. 2026a; Jiang et al. 2022b; Li et al. 2026], non-fixed joints are matched by Hungarian assignment, and we report a conjunctive success rate *j-succ.* that gates jointly on type, axis ($\leq$ 0.25 rad error), and origin ($\leq$ 3% of the local bounding-box diagonal); limit error is reported separately. The scale-relative origin gate avoids imposing one world-space tolerance on parts of very different sizes. Articulated-state geometry sweeps both objects through matched motion states and reports A-part CD and gIoU [Li et al. 2025]; reference-free articulation is summarised by SDF-based self-overlap under motion (A-overlap). Acceptance rules, full thresholds, and the worst-fill aggregation we use when a baseline cannot produce a prediction are in Supplementary Sec. G.

5.3 VLM-based Evaluation

Reference-based metrics require an annotated 3D target and are therefore unavailable on the In-the-Wild Suite; even with a target, a single RGB image often admits multiple valid 3D completions, especially for hidden parts and internal mechanisms. ArtiBench therefore complements reference-based metrics with a VLM-based protocol that applies to both suites. The judge (Gemini-3.1-Pro) receives the conditioning image together with diagnostic renders of the generated asset and scores five aspects on a 1–5 scale: part decomposition, part completeness, motion quality, joint type, and joint axis. The system prompt encodes mechanical priors with per-aspect rubrics; render templates and full prompts are in Supplementary Secs. G.5 and I.2.

We calibrate the judge with a human study using the same renders and rubrics. It attains sample-level Spearman $\rho \geq 0.80$ on every aspect, beating the strongest deterministic proxy by +0.05 to +0.21, and reaches system-level Spearman ≥ 0.93 on the per-method leaderboard (Supplementary Sec. G.6).

6 Experiments

6.1 Experimental Setup

Suites and inputs. We evaluate all methods on the two suites introduced in Section 5. The Category-Rich Suite contains 200 held-out ArtiAtlas objects with full 3D reference URDFs, while the In-the-Wild Suite contains 100 real photographs without unique 3D references and is evaluated with image-to-3D alignment, reference-free overlap, and calibrated VLM scores. We group methods by input contract: part-mesh-to-URDF methods receive reference part meshes, whole-mesh-to-URDF methods receive the assembled mesh, and image-to-URDF methods receive a rendered or real image.

Artisan and baselines. Artisan is trained on the ArtiAtlas training split and evaluated in the whole-mesh-to-URDF setting. It uses 32,768 input points during training and 102,400 at test time, with 64 part queries, decoder width 512, six bidirectional decoder layers, and eight attention heads. Approximately 1.5% of training assets exceed the 64-query capacity and are excluded from Artisan training. We train for 200,000 steps with AdamW, a learning rate of 10^{-4}, and a global batch size of 128 on 32 GPUs, using the same augmentation scheme as Particulate [Li et al. 2025]. At inference, queries are retained by a fixed present-probability times mask-quality threshold of 0.20. We run baselines with official implementations and released checkpoints when available; Real2Code is retrained because no weights are released. For the equal-data comparison in Tables 2 and 4, we also train Particulate on the ArtiAtlas training split for 200,000 steps, matching Artisan's training budget. For methods requiring auxiliary inputs, we provide ground-truth values on the

Table 2. **Category-Rich Suite results.** End-to-end articulation predictions evaluated against fully annotated 3D ground truth. Object geometry for image-input methods is reported separately in Table 3. *R-* and *A-* denote rest- and articulated-state metrics. The † rows apply connectivity refinement (Supplementary Sec. F.3); the ‡ rows first lift the image to a whole mesh before applying Artisan. Yellow columns are VLM-judged, blue columns are reference-free SDF-overlap metrics, and unshaded columns use paired 3D references. Gray dashes mark metrics not meaningful for the input modality.

Method	Part Decomposition					Articulation							
	R-part CD ↓	R-part gIoU ↑	decomp ↑	compl. ↑	R-overlap ↓	j-succ. ↑	limit err ↓	A-part CD ↓	A-part gIoU ↑	j-type ↑	j-axis ↑	motion ↑	A-overlap ↓
Part-mesh-to-URDF													
Real2Code	–	–	–	–	–	0.0156	2.74	0.359	0.136	2.79	2.15	1.32	0.104
ArtiPipe	–	–	–	–	–	**0.611**	**0.89**	**0.182**	**0.583**	**4.62**	**4.16**	**3.22**	**0.0854**
Articulate-Anything (PM)	–	–	–	–	–	0.172	1.82	0.478	0.265	4.09	3.24	2.06	0.106
Whole-mesh-to-URDF													
Particulate	0.270	0.226	2.35	3.08	0.0642	0.0824	2.46	0.390	0.0713	3.14	2.65	1.49	0.0611
Particulate†	0.285	0.257	2.65	3.20	0.151	0.0694	2.60	0.386	0.0821	2.90	2.69	1.85	0.127
Particulate (ArtiAtlas-trained)	0.225	0.362	2.93	3.78	0.0852	0.184	1.87	0.339	0.175	3.55	3.31	1.93	0.0657
Particulate (ArtiAtlas-trained)†	0.223	0.424	3.35	3.77	0.130	0.187	1.89	0.322	0.230	3.38	3.38	2.30	0.0962
URDF-Anything	0.376	0.0444	1.25	1.48	0.154	0.0307	3.02	0.694	-0.110	2.21	1.64	1.04	0.141
URDF-Anything†	0.413	0.0321	1.97	2.51	0.260	0.0299	3.02	0.715	-0.162	2.36	1.81	1.19	0.181
Artisan (ours)	**0.201**	0.427	3.43	**4.05**	**0.0484**	**0.297**	**1.86**	**0.291**	0.252	**3.67**	**3.65**	2.35	**0.0444**
Artisan (ours)†	0.208	**0.470**	**3.52**	3.92	0.0990	0.291	1.91	0.297	**0.270**	3.60	3.59	**2.54**	0.0767
Image-to-URDF													
PAct	0.319	-0.171	1.81	2.38	0.356	0.00798	3.03	0.432	-0.210	2.30	1.62	1.38	0.137
PhysX-3D	0.423	-0.127	1.24	1.36	0.468	0	3.01	0.511	-0.173	1.35	1.18	1.02	0.222
PhysX-Anything	0.433	-0.330	1.25	1.46	0.213	0.0187	3.03	0.491	-0.345	1.66	1.48	1.18	0.142
SINGAPO	0.371	-0.315	1.19	1.22	0.170	0.0192	3.04	0.487	-0.312	1.29	1.23	1.24	0.0752
URDFormer	0.452	-0.381	1.01	1.02	0.221	0.00600	3.16	0.535	-0.367	1.06	1.04	1.00	0.112
Articulate-Anything (I)	0.357	-0.291	2.58	2.94	0.123	0.0711	2.35	0.529	-0.356	3.23	2.51	1.69	0.0717
Artisan (ours)‡	**0.229**	**0.138**	2.92	3.83	0.0433	0.142	**2.11**	**0.287**	**0.0512**	**3.37**	3.33	**2.23**	0.0383
Artisan (ours)†‡	0.231	0.126	**3.17**	**3.85**	**0.0431**	**0.144**	2.11	0.289	0.0309	3.37	**3.40**	2.20	**0.0371**

Table 3. **Category-Rich Suite object geometry, image-input methods only.** Rest-state whole-object Chamfer distance against the 3D reference and image-to-shape alignment scores. Part-mesh- and whole-mesh-input methods are omitted because they receive the assembled geometry as input. Bold marks the best value per column.

Method	R-obj CD ↓	CLIP ↑	CLIP-N ↑	ULIP-2 ↑	Uni3D ↑
PAct	**0.0386**	0.678	0.608	0.346	0.312
PhysX-3D	0.0474	**0.853**	**0.666**	**0.415**	**0.383**
PhysX-Anything	0.0957	0.621	0.500	0.238	0.217
SINGAPO	0.0573	0.494	0.449	0.171	0.153
URDFormer	0.0835	0.455	0.421	0.172	0.128
Articulate-Anything (I)	0.0688	0.638	0.581	0.292	0.263

Category-Rich Suite and use the same VLM extraction procedure on the In-the-Wild Suite. Details are in the appendices.

6.2 Benchmark Results

Tables 2–4 summarize the main results. We report object geometry separately from articulation because several image-input methods first reconstruct or retrieve a complete shape before predicting functional structure.

Geometry versus articulation. On the Category-Rich Suite, PAct obtains the lowest whole-object Chamfer distance, while PhysX-3D achieves the strongest image-to-shape alignment scores (Table 3). However, these geometry scores do not translate into accurate articulation: image-to-URDF baselines remain far below mesh-input methods on part decomposition and joint success (Table 2). Lifting an image to a mesh and then applying Artisan (Artisan‡) recovers much of the decomposition and motion-score gap, but still trails the ground-truth-mesh setting, showing that residual shape errors affect downstream articulation.

Whole-mesh and part-mesh settings. Among whole-mesh-to-URDF methods, Artisan performs best. Compared with Particulate, it improves rest-state part Chamfer from 0.270 to 0.201, rest-state part gIoU from 0.226 to 0.427, and joint success from 0.0824 to 0.297. Training data and architecture both contribute to this gap: training Particulate on ArtiAtlas alone reaches 0.225, 0.362, and 0.184 on the same three metrics, and the remaining improvement comes from our architecture at matched training data and step budget. Connectivity refinement further improves decomposition quality, giving Artisan† the best whole-mesh rest-state gIoU (0.470) and decomposition score (3.52), but does not close the joint-success gap. The part-mesh-to-URDF setting provides an upper reference under oracle decomposition: ArtiPipe reaches 0.611 joint success, indicating that joint-structure and motion-parameter estimation remain difficult even after part ambiguity is removed. URDF-Anything, despite a larger language-model backbone, struggles on this broader suite, with low decomposition quality and joint success below 0.04.

In-the-wild evaluation. The In-the-Wild Suite shows the same trend without 3D ground truth (Table 4). Artisan achieves the highest whole-mesh decomposition and completeness scores, and its refined variant gives the best VLM-judged axis score. Training Particulate on ArtiAtlas lifts every VLM-judged dimension here as well, yet Artisan stays ahead on all five at matched training data. Articulate-Anything obtains strong joint-type scores among image-input methods, suggesting useful image-conditioned priors, but remains weaker on axis and motion quality. PhysX-3D again leads

Table 4. **In-the-Wild Suite results.** Image-conditioned articulation predictions on real photographs without 3D ground truth. Shape fidelity is measured by image-to-3D alignment; decomposition and motion are judged by a calibrated VLM; self-overlap is measured by SDF intersection.

Method	Object Geometry				Part Decomposition			Articulation			
	CLIP ↑	CLIP-N ↑	ULIP-2 ↑	Uni3D ↑	decomp ↑	compl. ↑	R-overlap ↓	j-type ↑	j-axis ↑	motion ↑	A-overlap ↓
Whole-mesh-to-URDF											
Particulate	–	–	–	–	1.95	2.62	0.0155	3.14	2.64	1.63	**0.0164**
Particulate†	–	–	–	–	2.45	3.05	0.0151	3.07	2.81	1.72	0.0259
Particulate (ArtiAtlas-trained)	–	–	–	–	2.33	3.45	**0.0136**	3.51	3.34	1.74	0.0358
Particulate (ArtiAtlas-trained)†	–	–	–	–	2.79	3.65	0.0286	3.53	3.36	1.82	0.0459
URDF-Anything	–	–	–	–	1.15	1.35	0.0916	2.44	1.49	1.05	0.0887
URDF-Anything†	–	–	–	–	1.34	1.61	0.0916	2.23	1.38	1.04	0.0904
Artisan (ours)	–	–	–	–	2.88	**4.09**	0.0143	**3.57**	3.62	**2.08**	0.0224
Artisan (ours)†	–	–	–	–	**3.05**	3.94	0.0155	3.55	**3.65**	1.99	0.0235
Image-to-URDF											
PAct	0.578	0.523	0.321	0.288	1.60	2.00	0.383	2.16	1.56	1.29	0.102
PhysX-3D	**0.770**	**0.584**	**0.385**	**0.352**	1.22	1.34	0.462	1.32	1.17	1.06	0.139
PhysX-Anything	0.532	0.411	0.207	0.181	1.29	1.69	**0.130**	1.75	1.61	1.15	**0.0767**
SINGAPO	0.348	0.359	0.131	0.126	1.16	1.29	0.247	1.31	1.28	1.18	0.0937
URDFormer	0.321	0.335	0.122	0.113	1.04	1.09	0.231	1.11	1.10	1.06	0.110
Articulate-Anything (I)	0.476	0.471	0.241	0.231	**2.58**	**3.17**	0.190	**3.36**	**2.40**	**1.60**	0.0932

Table 5. **Model ablation on the Category-Rich Suite.** All variants are evaluated at 100K training steps. Metric definitions follow Table 2. D1 denotes mask-aware bidirectional decoding; D2 denotes the training-only axis-pivot loss; Q1 denotes canonical axis-sign/range parameterization.

Variant	R-gIoU↑	decomp↑	j-succ.↑	A-gIoU↑	j-axis↑	motion↑
Particulate re-trained	0.359	2.88	0.168	0.170	3.28	1.86
Ours vanilla	0.383	3.00	0.179	0.192	3.30	2.03
+ Mask-aware decoder (D1)	0.398	3.18	0.227	0.214	3.42	2.10
+ Axis-pivot loss (D2)	0.376	3.01	0.221	0.193	3.38	2.05
+ D1 + D2	0.400	3.16	0.234	0.227	3.48	2.22
+ Canonical parameterization (Q1)	**0.415**	**3.37**	**0.266**	**0.231**	**3.57**	**2.30**

image-to-shape alignment while producing weak articulation, reinforcing the gap between visual fidelity and functional structure.

6.3 Ablation Studies

Table 5 isolates model components under the same training data, inference policy, and Category-Rich Suite evaluation. All rows are evaluated at 100K steps and should be read as controlled comparisons rather than final headline numbers. Retraining a Particulate-style baseline gives 0.168 joint success. Our query interface and bidirectional backbone raise it to 0.179; mask-aware decoding gives the largest gain, reaching 0.227; and the axis-pivot loss further improves joint success to 0.234 when combined with mask-aware decoding. Canonical motion parameterization gives the strongest ablation row, with 0.266 joint success, 0.415 rest-state gIoU, and the best axis and motion scores.

7 Conclusion

We present ArtiAtlas, ArtiPipe, Artisan, and ArtiBench as coupled infrastructure for diverse 3D articulation modeling. Our experiments show that static shape quality alone does not imply accurate articulation: current methods still struggle with part decomposition, kinematic structure, joint axes, and motion limits, especially on long-tail categories and complex mechanisms. These results suggest that progress requires scalable articulated-object data, models designed for variable-cardinality structure, and evaluation protocols that measure both geometry and motion validity. Our work also has limitations. ArtiAtlas may contain residual annotation noise, and VLM- or image-generation-assisted stages may introduce biases. The In-the-Wild Suite does not cover the full diversity of real-world objects, and both suites prioritize category breadth over sufficient repeated examples for robust per-category estimates. Our trained model focuses on the whole-mesh-to-URDF setting. Current metrics assess individual joints and state-sampled overlap rather than functional multi-joint operation, and the VLM judge assesses diagnostic renders rather than simulator behavior. Finally, predicted URDFs are geometric and kinematic hypotheses rather than certified physically executable mechanisms. We plan to release the 10,005-object public subset of ArtiAtlas, benchmark splits, evaluation code, trained Artisan checkpoints, and all redistributable annotations. For assets restricted by license, we will provide metadata, permitted derived annotations, and reconstruction instructions when possible. We hope this infrastructure makes the remaining gaps in diverse articulation modeling measurable and supports future work on interactable 3D assets.

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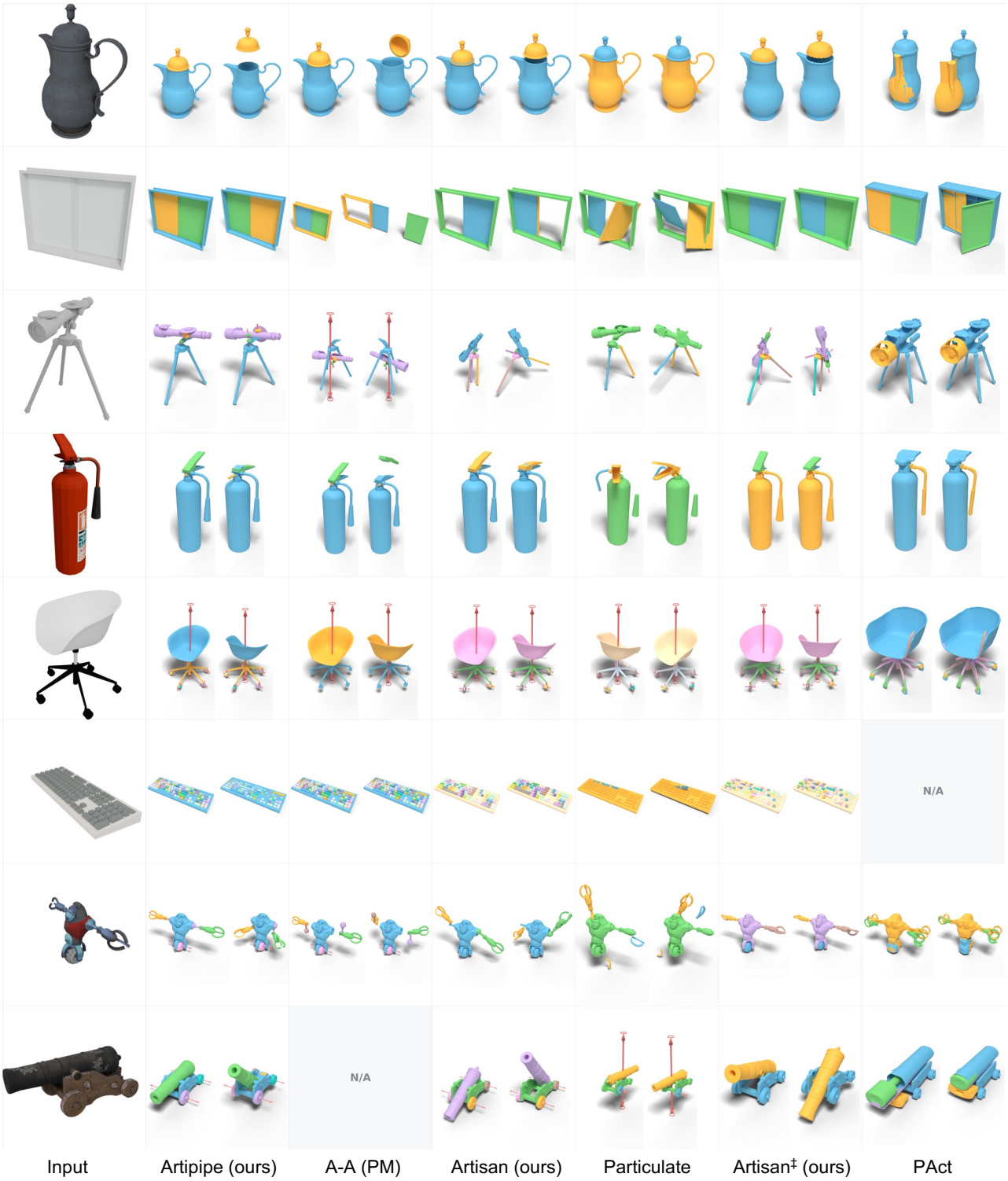


Fig. 4. **Qualitative comparison on the Category-Rich Suite.** Each row shows one held-out asset of ArtiAtlas; columns are the input mesh followed by six methods, with two renders per method showing the assembled URDF driven to its joints' lower limits (left) and upper limits (right). ArtiPipe is our annotation pipeline operating on GT part meshes; Artisan is our whole-mesh-to-URDF model; the ‡ column first lifts the conditioning image to a whole mesh with Seed3D 2.0 and then runs Artisan on the synthesised mesh. A-A: Articulate-Anything. *N/A* marks cells for which a method cannot produce a result. Rows are chosen to expose repeated parts, hidden chains, small moving components, and long-tail categories outside PartNet-Mobility.

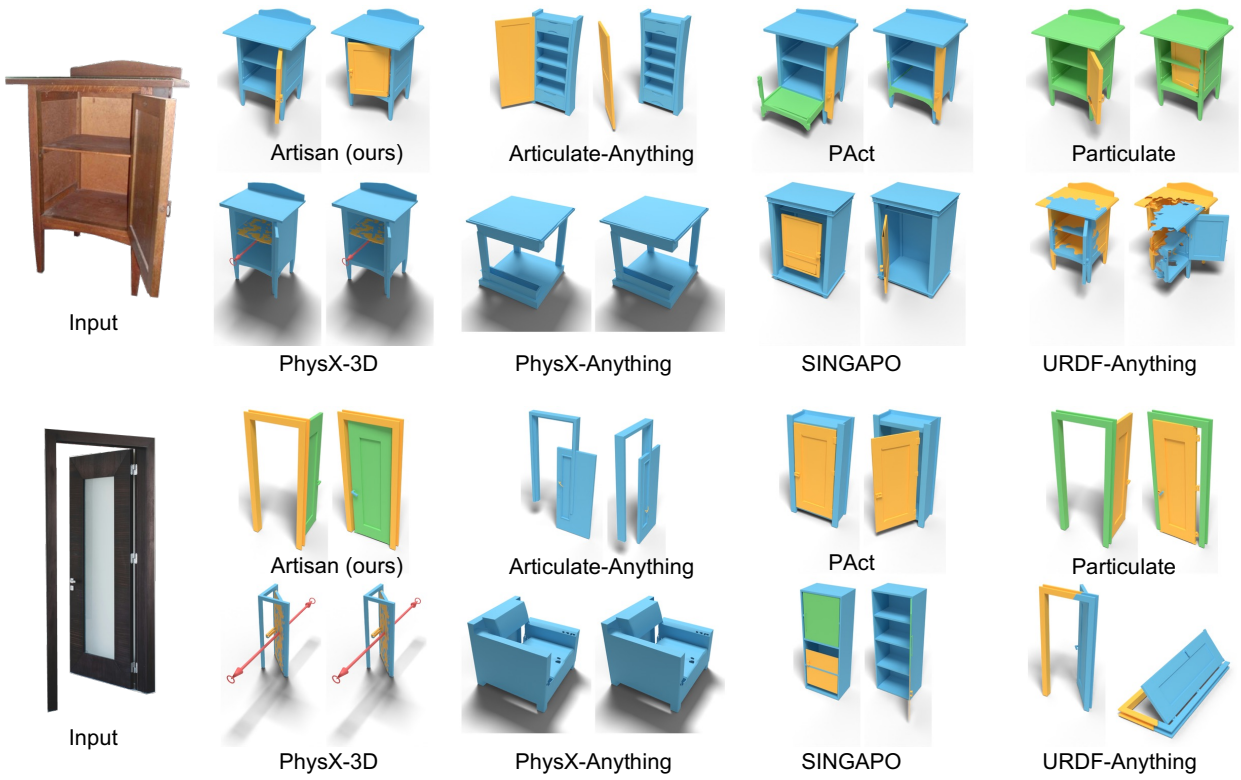


Fig. 5. **Qualitative comparison on the In-the-Wild Suite.** Each block shows one real photograph used as the conditioning input on the left, followed by the URDF predictions from eight methods arranged in a 2×4 grid. Each method tile shows the assembled URDF driven to its joints' lower limits (left) and upper limits (right). The In-the-Wild Suite contains no 3D ground truth, so image-input methods consume the photograph directly, whole-mesh-to-URDF methods consume the foreground mesh recovered from the photograph by Seed3D 2.0, and Artisan is reported in the whole-mesh setting. Examples are chosen to expose hidden articulation chains and categories outside PartNet-Mobility.



Fig. 6. **Representative Artisan predictions and remaining failure modes.** Each cell shows one asset driven to its joints' lower limits (left) and upper limits (right). Rows 1–2 sample In-the-Wild assets where Artisan recovers a usable kinematic chain from the photograph-derived foreground mesh. Row 3 samples Category-Rich Suite assets that expose two recurring failure modes: small movable parts are sometimes under-segmented, as in the turntable casing whose control buttons are absent or the piano whose individual keys are merged into a single rigid block; and motion ranges are occasionally over-predicted, driving self-penetration between adjacent links at the upper-limit pose.